

Chuyên đề Lập trình ứng dụng Web

HTML – CSS – JavaScript

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- ◆ HTML DOM
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References

- ◆ HTML: <https://www.w3schools.com/html/>
- ◆ CSS: <https://www.w3schools.com/css/>
- ◆ JavaScript: <https://www.w3schools.com/js/>

What is DOM?

◆ The Document Object Model (DOM)

- DOM represents a document as a *tree* structure, where
- Each node is an object representing a part of the document.
- **Nodes:**
 - **Document Node:** The *root node* that represents the entire document.
 - **Element Node:** Represents HTML or XML elements.
 - **Attribute Node:** Represents attributes of HTML or XML elements.
 - **Text Node:** Represents the text content inside elements

```
<!DOCTYPE html>
<html>
  <body>
    <h1 style="color: ■ red;">Hello, World!</h1>
  </body>
</html>
```

DOM tree

```
Document
├── html
│   ├── head
│   │   └── title
│   │       └── "Example"
│   └── body
│       ├── h1
│       │   └── "Hello, World!"
│       └── p
│           └── "This is a paragraph."
```

```
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Example</title>
5      </head>
6      <body>
7          <h1>Hello, World!</h1>
8          <p>This is a paragraph.</p>
9      </body>
10 </html>
```

HTML Elements

- ◆ `<tagname>`Content goes here...`</tagname>`
- ◆ Nested HTML Elements

```
<!DOCTYPE html>  
<html>  
  <body>  
  
    <h1>My First Heading</h1>  
    <p>My first paragraph.</p>  
  
  </body>  
</html>
```

Accessing DOM Elements (1)

- ◆ When an HTML document is loaded into a web browser, it becomes a **document object**.
- ◆ The document object is the **root node** of the HTML document.
- ◆ The document object is a **property** of the window object
 - `window.document` or `document`
 - `let url = window.document.URL;`
 - `let url = document.URL;`

Accessing DOM Elements (2)

◆ Using JavaScript

```
// Access element by ID
var element = document.getElementById('myElement');

// Access elements by class name
var elements = document.getElementsByClassName('myClass');

// Access element using a CSS selector
var element = document.querySelector('.myClass');

// Access all elements matching a CSS selector
var elements = document.querySelectorAll('.myClass');
```

```
// Change the content of an element
element.innerHTML = 'New Content';

// Change the attributes of an element
element.setAttribute('src', 'newImage.png');
```

```
// Create a new element
var newElement = document.createElement('div');
newElement.innerHTML = 'This is a new element';

// Add the new element to the DOM
document.body.appendChild(newElement);

// Remove an element from the DOM
document.body.removeChild(newElement);
```

What is CSS?

- ◆ **C**a**s**ca**d**i**n**g **S**t**y**e **S**h**e**e**t**s (CSS) is used to format the layout of a webpage.
 - color, font, the size of text, the spacing between elements,...
- ◆ **Cascading**
 - a style applied to a parent element will also apply to all children elements within the parent

Using CSS

- ◆ **Inline** - by using the style attribute inside HTML elements

- `<h1 style="color:blue;">A Blue Heading</h1>`
`<p style="color:red;">A red paragraph.</p>`

- ◆ **Internal** - by using a `<style>` element in the `<head>` section

- `<style>`
 `body {background-color: powderblue;}`
 `h1 {color: blue;}`
 `p {color: red;}`
`</style>`

- ◆ **External** - by using a `<link>` element to link to an external CSS file

- `<link rel="stylesheet" href="styles.css">`

HTML Forms

- ◆ An HTML form is used to collect user input.
- ◆ The user input is most often sent to a server for processing.

First name:

Last name:

```
<form>  
.  
form elements  
.  
</form>
```

```
<form>  
  <label for="fname">First name:</label><br>  
  <input type="text" id="fname" name="fname"><br>  
  <label for="lname">Last name:</label><br>  
  <input type="text" id="lname" name="lname">  
</form>
```

The <label> Element

- ◆ The **for** attribute of the <label> tag should be **equal** to the **id** attribute of the <input> element to bind them together.

Choose your favorite Web language:

- ☐ HTML
- ☐ CSS
- ☐ JavaScript

```
<p>Choose your favorite Web language:</p>
```

```
<form>
```

```
  <input type="radio" id="html" name="fav_language" value="HTML">
```

```
  <label for="html">HTML</label><br>
```

```
  <input type="radio" id="css" name="fav_language" value="CSS">
```

```
  <label for="css">CSS</label><br>
```

```
  <input type="radio" id="javascript" name="fav_language" value="JavaScript">
```

```
  <label for="javascript">JavaScript</label>
```

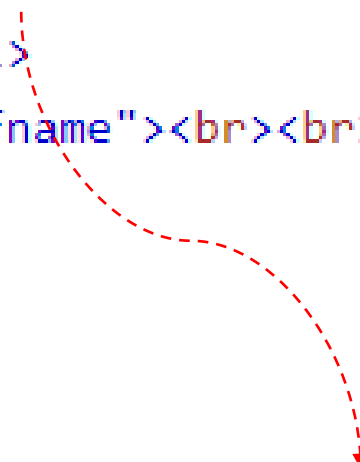
```
</form>
```

The *form* Attribute

- ◆ An input field located **outside** of the HTML form but still a part of the form

```
<form action="/action_page.php" id="form1">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname"><br><br>
  <input type="submit" value="Submit">
</form>

<label for="lname">Last name:</label>
<input type="text" id="lname" name="lname" form="form1">
```



The *formaction* Attribute

- ◆ An HTML form with two submit buttons, with different actions

```
<form action="/action_page.php">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname"><br><br>
  <label for="lname">Last name:</label>
  <input type="text" id="lname" name="lname"><br><br>
  <input type="submit" value="Submit">
  <input type="submit" formaction="/action_page2.php" value="Submit as Admin">
</form>
```

The *formmethod* Attribute

- ◆ A form with two submit buttons with "get" and "post" method

```
<form action="/action_page.php" method="get">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname"><br><br>
  <label for="lname">Last name:</label>
  <input type="text" id="lname" name="lname"><br><br>
  <input type="submit" value="Submit using GET">
  <input type="submit" formmethod="post" value="Submit using POST">
</form>
```

HTML DOM Events

- ◆ DOM Events allow JavaScript to add **event listener** or **event handlers** to HTML elements.
 - HTML onclick
 - `<button onclick="myFunction()">Click me</button>`
 - JavaScript click
 - `button.addEventListener("click", myFunction);`

HTML JavaScript

◆ Why JavaScript?

- **HTML** to define the *content* of web pages
- **CSS** to specify the *layout* of web pages
- **JavaScript** to program the *behavior* of web pages

◆ Version of JavaScript

- The Original JavaScript ES1 ES2 ES3 (1997-1999)
- The First Main Revision ES5 (2009)
- The Second Revision ES6 (2015)
- The Yearly Additions (2016, 2017 ... 2021, 2022)

JavaScript Can Change HTML Styles (CSS)

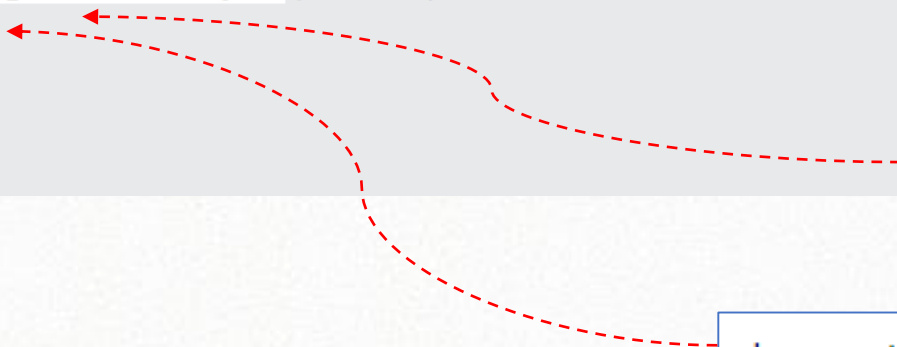
- ◆ Using the HTML `<script>` tag to define a client-side script (JavaScript).

```
<body>

<p id="demo">Hi.</p>

<script>
document.getElementById("demo").innerHTML = "Hello World!";
</script>

</body>
```



```
document.getElementById("demo").style.fontSize = "25px";
document.getElementById("demo").style.color = "red";
document.getElementById("demo").style.backgroundColor = "yellow";
```

```
document.getElementById("image").src = "picture.gif";
```

JavaScript Functions

- ◆ JavaScript in <head>
- ◆ JavaScript in <body>

```
<p id="demo">A Paragraph</p>

<button type="button" onclick="myFunction()">Try it</button>

<script>
function myFunction() {
    document.getElementById("demo").innerHTML = "Paragraph changed.";
}
</script>
```

External JavaScript

- ◆ External file *myScript.js*

```
function myFunction() {  
    document.getElementById("demo").innerHTML = "Paragraph changed.";  
}
```

- ◆ Using an external script

```
<script src="myScript.js"></script>  
  
<script src="/js/myScript1.js"></script>  
<script src="/js/myScript2.js"></script>
```

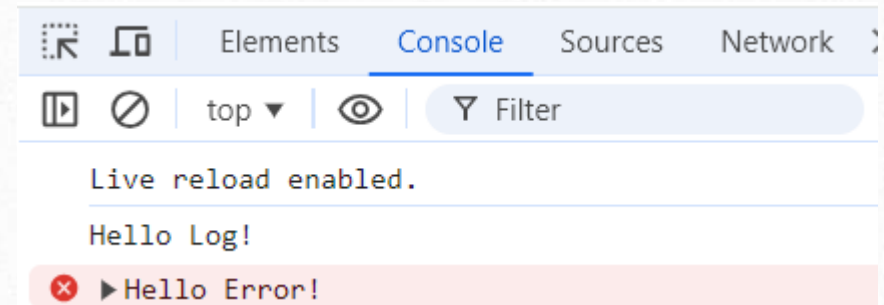
- ◆ External References

```
<script src="https://www.w3schools.com/js/myScript.js"></script>
```

The Window Object

- ◆ The window object represents an open window in a browser.
 - Window alert()
 - Window confirm()
 - Window Console Object: access to the browser's debugging console.

```
window.console.error("You made a mistake");  
console.error("You made a mistake");  
console.log("Hello World!");
```



Web APIs

- ◆ API stands for Application Programming Interface.
- ◆ A **Web API** is an application programming interface for the *Web*.
- ◆ A **Browser API** can extend the functionality of a *web browser*.
- ◆ A **Server API** can extend the functionality of a *web server*.

Browser APIs

- ◆ All browsers have a set of built-in Web APIs to support complex operations, and to help accessing data.
- ◆ For example, the Geolocation API can return the coordinates of where the browser is located.

Third Party APIs

- ◆ Third party APIs are not built into your browser.
- ◆ To use these APIs, you will have to download the code from the Web.
- ◆ Examples:
 - YouTube API - Allows you to display videos on a web site.
 - Twitter API - Allows you to display Tweets on a web site.
 - Facebook API - Allows you to display Facebook info on a web site.

Exercise

- ◆ Xây dựng trang html có sử dụng
 - External CSS
 - Form và Events
 - External JavaScript, sử dụng function để thay đổi CSS, thay đổi elements của form
 - `console.log` và `console.error`
 - Web APIs (có sẵn trong trình duyệt) hoặc third-party APIs